Output Form (OF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *MMLU* | ***Context:*** *North India Competitive Exam Aspirants (Hindi-Medium)*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is the dimension with the strongest partial match — both MMLU and the deployment use text-based MCQ as the interaction vehicle, and the binary correct/incorrect verdict can in principle be derived from MMLU's A/B/C/D output [Q83]. However, MMLU evaluates only single-letter classification accuracy [Q34, Q41]; it does not evaluate any natural-language rationale, Hindi fluency, or pedagogical appropriateness. Calibration failures further reduce the value of MMLU's confidence outputs as deployment signal: GPT-3 confidence deviates from accuracy by up to 24% zero-shot [Q57] and 14% few-shot [Q96], with Elementary Mathematics RMS error of 19.4% [Q58]. Given the deployment's negative-marking exam context where confidence calibration matters, this is a real concern. The MODERATE priority weight from elicitation reflects this partial match — score is 2 rather than 1 because the structural MCQ alignment provides some external validity.

ID	Evidence
Q34	“To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks.” (p.5)
Q41	“All values are percentages.” (p.5)
Q57	“GPT-3 is uncalibrated. In fact, its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects.” (p.7)
Q58	“Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019). Many tasks have miscalibrated predictions, such as Elementary Mathematics which has a zero-shot RMS calibration error of 19.4%.” (p.7)
Q83	“Models are fine-tuned to predict one of four classes using the UnifiedQA MCQ questions and using our dev+val set. We test on our multitask test set.” (p.12)
Q96	“While models are more calibrated in a few-shot setting than a zero-shot setting, they are still miscalibrated, with gap between accuracy and confidence reaching up to 14%. Here the correlation between confidence and accuracy is r = 0.81, compared to r = 0.63 in the zero-shot setting.” (p.13)

Strengths

- Text-based MCQ output form is a partial match to deployment interaction vehicle [Q44, DATASET strength 1].
- Balanced label distribution across A/B/C/D classes prevents position-bias artifacts [DATASET strength 3].
- Confidence/calibration measurement framework [Q56, Q58] is methodologically transferable, even if the specific MMLU calibration findings are not directly applicable.

ID	Evidence
Q44	“We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject.” (p.6)
Q56	“We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject.” (p.7)
Q58	“Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019). Many tasks have miscalibrated predictions, such as Elementary Mathematics which has a zero-shot RMS calibration error of 19.4%.” (p.7)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Partial match — MCQ text output is appropriate, but the deployment also requires generative Hindi explanatory text which MMLU does not evaluate.

OF-2: Not applicable — deployment is text-based, not speech-based; TTS availability is not a primary requirement per elicitation.

OF-3: Literacy is not a barrier for the deployment population (graduate-level aspirants per region YAML); however, mobile-first low-bandwidth constraints apply [WEB-10] and rural shared-device usage is documented [WEB-12].

OF-4: External validity is partially harmed: a model's MMLU label-accuracy and calibration scores do not predict its capacity to produce coherent Hindi explanatory output, and calibration failures up to 24% [Q57] mean confidence signals do not transfer reliably.

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Q34	“To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks.” (p.5)
Q57	“GPT-3 is uncalibrated. In fact, its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects.” (p.7)
Q96	“While models are more calibrated in a few-shot setting than a zero-shot setting, they are still miscalibrated, with gap between accuracy and confidence reaching up to 14%. Here the correlation between confidence and accuracy is r = 0.81, compared to r = 0.63 in the zero-shot setting.” (p.13)
WEB-10	IAMAI/Kantar 2024: Bihar 43%, UP 46% internet penetration — relevant for mobile-first deployment constraints (https://www.iamai.in/sites/default/files/research/Kantar_%20IAMAI%20report_2024_.pdf)
WEB-12	https://www.business-standard.com/industry/news/india-to-exceed-900mn-internet-users-by-2025-125011600669_1.html
WEB-21	No Hindi exam-prep explanation-quality rubric exists (https://arxiv.org/abs/2508.19831)

Information Gaps

- How well MMLU's calibration framework translates to a Hindi-output context is not empirically tested.

Requires Expert Verification

- Mobile-device performance of any MMLU-evaluated model on the deployment hardware profile (mid-range Android, intermittent connectivity).

Remediation

Gap 1: Calibration failures (up to 24% off [Q57]) reduce reliability of confidence signals in a negative-marking exam context.

Recommendation: Add explicit calibration measurement on Hindi-medium UPSC/SSC items (e.g., RMS calibration error per subject) using the methodology from MMLU [Q58] but applied to Indian-curriculum benchmarks; require model abstention support for negative-marking scenarios.

ID	Evidence

Q57	“GPT-3 is uncalibrated. In fact, its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects.” (p.7)
Q58	“Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019). Many tasks have miscalibrated predictions, such as Elementary Mathematics which has a zero-shot RMS calibration error of 19.4%.” (p.7)